UTS Mini Mixing Desk — Compressor — one sheet per section

1 Connector

File: connector.kicad_sch

2 Input

File: input.kicad_sch

3 VCA

File: vca.kicad_sch

4 Output

File: output.kicad_sch

5 Sidechain

File: sidechain.kicad_sch

6 Power

File: power.kicad_sch

7 Meters

File: meters.kicad_sch

```
.param LEVEL_DBU=4 FREQ=1k
.param THRESHOLD=0.5 RATIO=1 ATTACK=0.5 RELEASE=0.5 MAKEUP=0
.param TRIM=0.5 GRTRIM=0.5 LVLTRIM=0.5
.param BYPASS=0 KEY=0 HPF_DEFEAT=0 LINK=0
* Test bench (Inspect > Simulator): see sim/README.md. Knobs run 0 to 1.
* Rack supply and ground. AGND is SPICE ground 0.
VGND AGND 0 0
VPOS +16V-IN 0 16
VNEG -16V-IN 0 -16
RPGND PGND 0 0.01
* Balanced source, 50R per leg, LEVEL_DBU differential
.param VPK={0.7746*sqrt(2)*pwr(10,LEVEL_DBU/20)}
VINP SRC+ 0 DC 0 AC 0.5 SIN(0 {VPK/2} {FREQ})
VINN SRC- 0 DC 0 AC 0.5 180 SIN(0 {-VPK/2} {FREQ})
RSP SRC+ IN+ 50
RSN SRC- IN- 50
* 20k balanced load; aux inputs and link bus open
RLP OUT+ 0 10k
RLN OUT- 0 10k
RAUXP AUXIN+ 0 1G
RAUXN AUXIN- 0 1G
RLINK LINK 0 1G
EOUT OUT_DIFF 0 OUT+ OUT- 1
ROUT OUT_DIFF 0 1G
* Solver: the 1G shunt on every node stops ngspice stalling while it compresses hard
.options method=gear rshunt=1e9
```

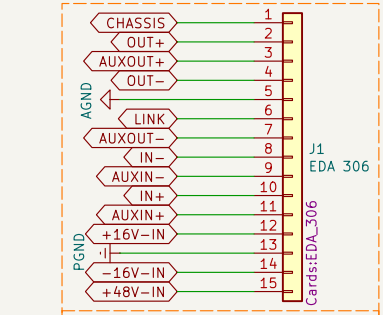
UTS Mini Mixing Desk

Sheet: /

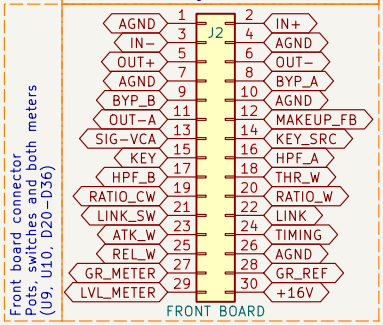
File: UTS Mini Mixing Desk — Compressor.kicad_sch

Title: UTS Mini Mixing Desk — Compressor

Size: A4	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/8



On board copper female connector.
15 Poles. Labeling from Midas.



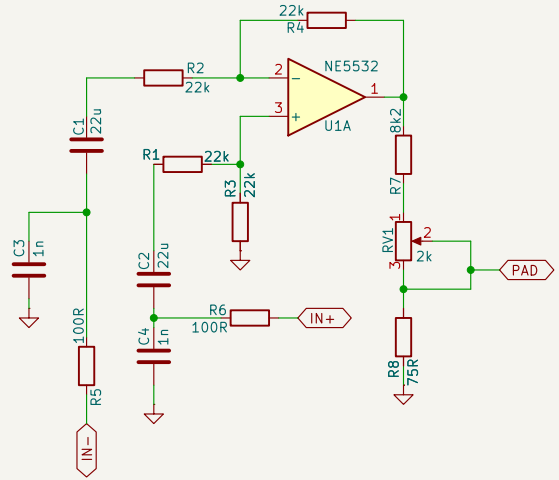
UTS Mini Mixing Desk

Sheet: /1 Connector/
File: connector.kicad_sch

Title: UTS Mini Mixing Desk — Compressor

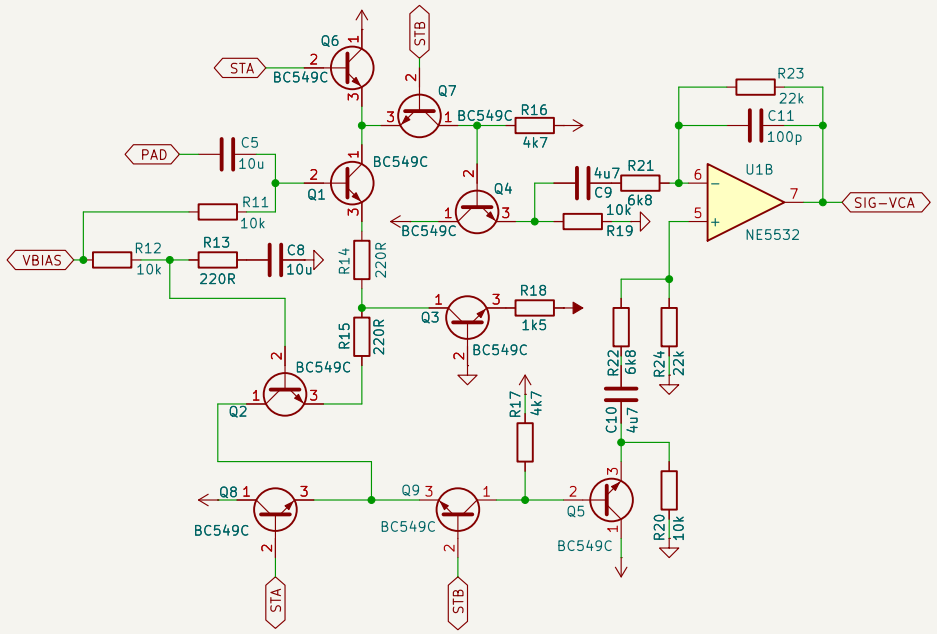
Size: User	Date: 2026−08−31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/8

Input Section
Buffer and Unbalance the
Input Before Sending it to the Pad



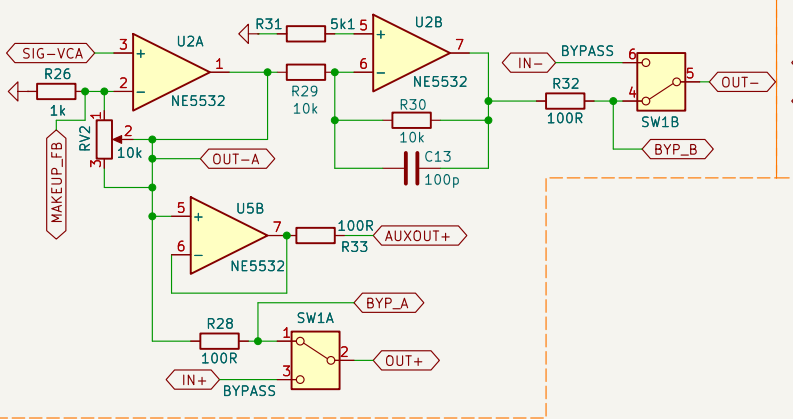
UTS Mini Mixing Desk		
Sheet: /2 Input/		
File: input.kicad_sch		
Title: UTS Mini Mixing Desk – Compressor		
Size: User	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/8

The Gain Cell
Nine Transistors & an Op-Amp to act as a Voltage Controlled Amplifier

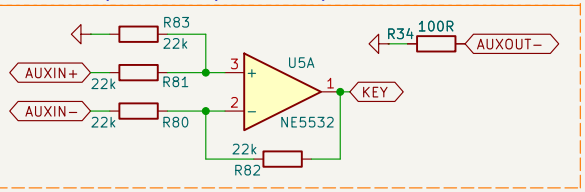


UTS Mini Mixing Desk		
Sheet: /3 VCA/		
File: vca.kicad_sch		
Title: UTS Mini Mixing Desk – Compressor		
Size: User	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/8

Output Chain, Buffer Op-Amps and Signal Balancing



Aux Output & Input Setup



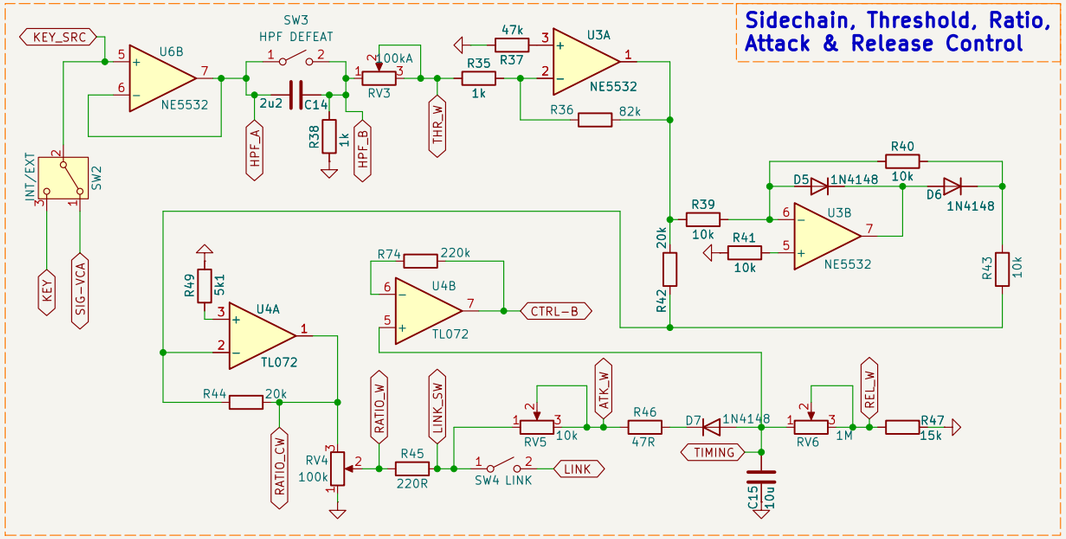
UTS Mini Mixing Desk

Sheet: /4 Output/

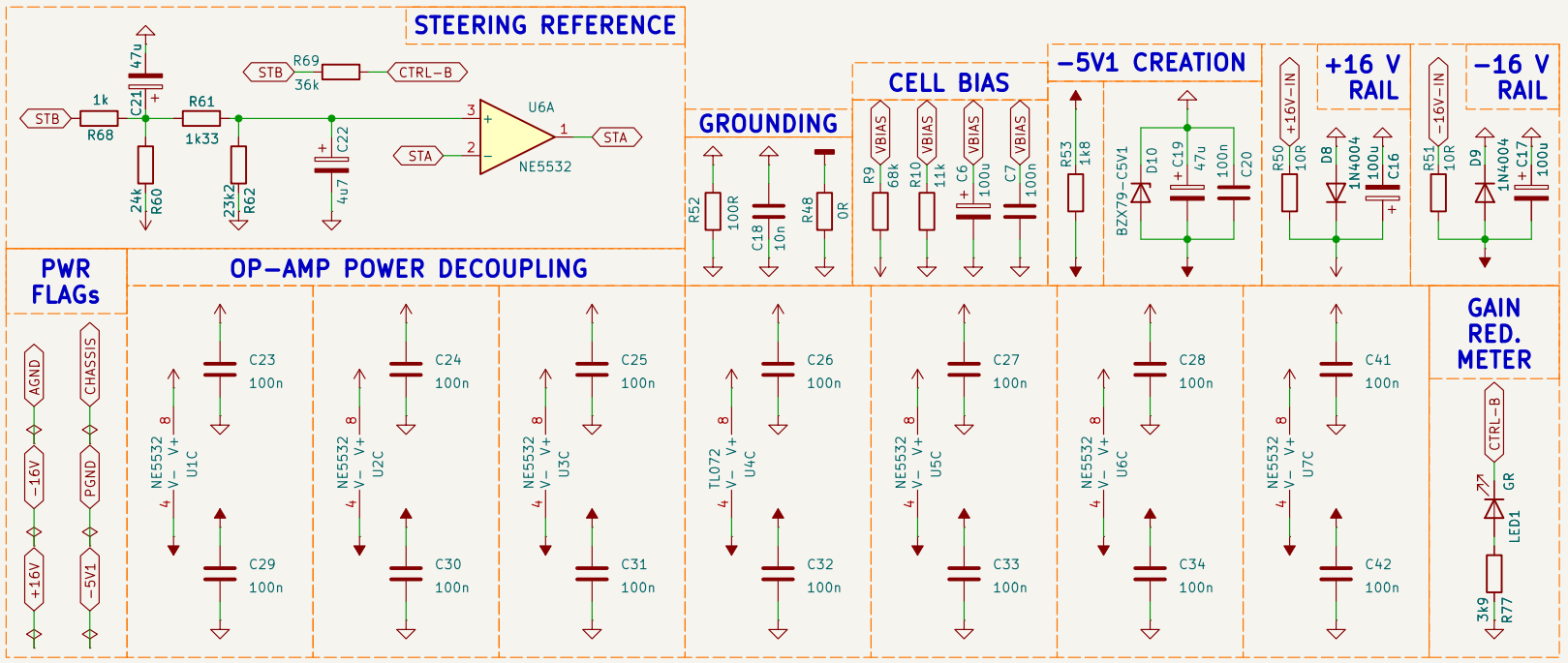
File: output.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

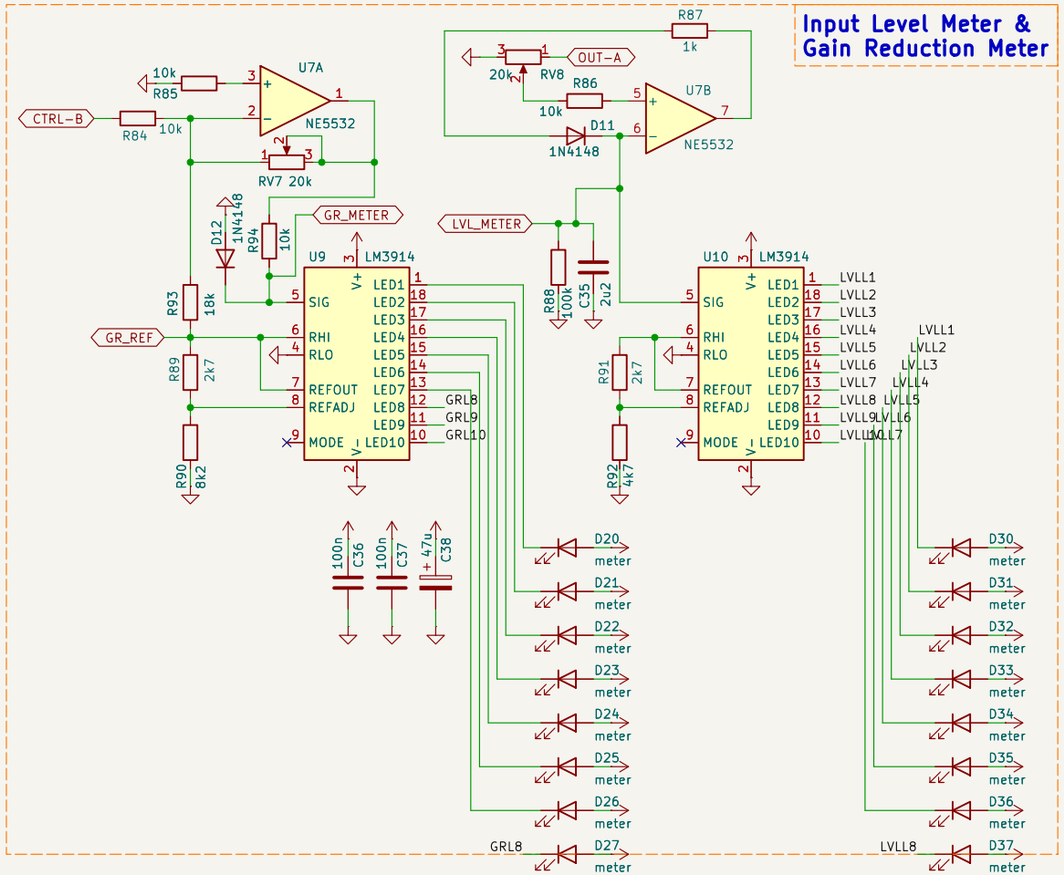
Size: User	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/8



UTS Mini Mixing Desk		
Sheet: /5 Sidechain/		
File: sidechain.kicad_sch		
Title: UTS Mini Mixing Desk — Compressor		
Size: User	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/8



UTS Mini Mixing Desk		
Sheet: /6 Power/		
File: power.kicad_sch		
Title: UTS Mini Mixing Desk – Compressor		
Size: User	Date: 2026-08-31	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/8



UTS Mini Mixing Desk

Sheet: /7 Meters/

File: meters.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

Size: User	Date: 2026-09-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 8/8